MOHAMMED ALGHANAM

COMPUTER SCIENCE STUDENT | DATA SCIENCE & ANALYTICS

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PROFILE

Computer Science student at Majmaah University seeking cooperative training in data science, data analytics, or related fields. Hands-on academic experience building a text-based recommendation system using TF-IDF and cosine similarity, and developing Python tools for numerical analysis. Familiar with Python, SQL, and data mining concepts; eager to apply analytical thinking and strengthen practical skills through real-world data projects.

EDUCATION

Majmaah University | Bachelor's Degree Student - Computer Science / Data Science

Relevant Coursework: Data Mining, Artificial Intelligence, Numerical Analysis, Parallel Computing, Software Modeling, Compiler Design

TECHNICAL SKILLS

Data & Analytical Methods: TF-IDF, cosine similarity, content-based recommendation, numerical interpolation

Programming: Python, SQL, MATLAB, C++, Prolog

Frameworks & APIs: Flask, SQLAlchemy, Groq API, SambaNova API

Tools & Other: Git, Linux command line, OpenMP, draw.io, VirtualBox

SELECTED ACADEMIC PROJECTS

Al-Warraq - Academic Recommendation & Search Web App | Python, Flask, SQLAlchemy

- Built a content-based recommendation and search feature using TF-IDF and cosine similarity to rank academic materials by relevance.
- Developed database-backed functionality with Flask and SQLAlchemy to support the web application and recommendations.
- Integrated Groq and SambaNova language-model APIs with fallback handling, basic request limits, and prompt-injection safeguards.

Numerical Analysis GUI | Python

- Developed a graphical tool that generates Newton divided-difference tables, constructs interpolation polynomials, and calculates roots.

Analysis and Data Visualization of Traffic Accident in Saudi Arabia | R

- Analyzed traffic accident data in Saudi Arabia using R to identify patterns and trends in accident rates.
- Performed statistical analysis and created data visualizations to present findings and highlight road safety trends.

ADDITIONAL ACADEMIC EXPERIENCE

- Compared Saudi Arabia's PDPL and the EU GDPR in an academic analysis of personal data privacy requirements.
- Completed parallel-programming exercises involving multithreading and reductions with C++ and OpenMP.